

Qualification Round Problem Set (Sample)

Five reasoning problems, A through E. Solve them, then submit your answers on the Submission page.

Problem A (Deductive, 5 pts)

Three friends — Ada, Ben, and Cleo — each make one statement. Ada says "Ben is lying." Ben says "Cleo is lying." Cleo says "Ada and Ben are both lying." Exactly one of the three is telling the truth. Who is it, and why?

Problem B (Probabilistic, 5 pts)

A rare disease affects 1 in 500 people. A test for it is 98% accurate for both true positives and true negatives. A random person tests positive. What is the probability they actually have the disease? Show your reasoning, not just the number.

Problem C (Causal, 5 pts)

A city notices that neighborhoods with more ice cream shops also have higher rates of sunburn. A local paper concludes ice cream causes sunburn. Identify the likely confound and explain how you would test whether the relationship is causal.

Problem D (Strategic, 5 pts)

Two vendors set prices for an identical product without seeing each other's price first. Higher combined sales go to whichever price is lower; if tied, sales split evenly. Describe the equilibrium you would expect and why undercutting doesn't spiral indefinitely in practice.

Problem E (Estimation, 5 pts)

Estimate how many students worldwide will take a competitive reasoning exam of some kind this year. State your assumptions explicitly and show the calculation — we grade the reasoning, not the exact figure.

How to submit

Work through all five problems on your own paper or document. When ready, go to the Submission page, fill in your details, and paste or summarize your answers in the answers field before the deadline.